

① OSI MODEL (7 LAYERS)

7 Application	Network apps
6 Presentation	Data format, encryption
5 Session	Session management
4 Transport	End-to-end, ports
3 Network	Logical addressing, routing
2 Data Link	MAC addressing, switching
1 Physical	Bits, cables, signals

② TCP/IP MODEL (4 LAYERS)

Application	HTTP, DNS, DHCP, etc.
Transport	TCP, UDP
Internet	IP, ICMP
Network Access	Ethernet, Wi-Fi

③ COMMON PORTS

- 20/21 - FTP
- 22 - SSH
- 23 - Telnet
- 25 - SMTP
- 53 - DNS
- 67/68 - DHCP
- 80 - HTTP
- 110 - POP3
- 143 - IMAP
- 389 - LDAP
- 443 - HTTPS
- 161/162 - SNMP

④ IP ADDRESSING

IPv4: 32-bit address
192.168.1.1

Subnet Mask: 255.255.255.0 /24

Private Ranges:

- 10.0.0.0 - 10.255.255.255
- 172.16.0.0 - 172.31.255.255
- 192.168.0.0 - 192.168.255.255

Loopback: 127.0.0.1

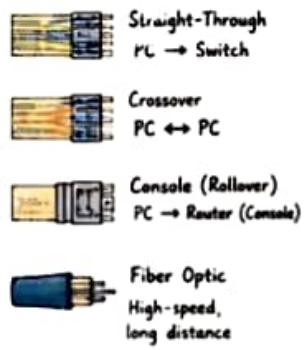
⑤ SUBNETTING QUICK REFERENCE

Prefix	Mask	Usable Hosts
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14
/29	255.255.255.248	6
/30	255.255.255.252	2

⑥ DEVICE TYPES



⑦ CABLE TYPES



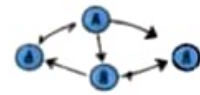
⑧ COMMON COMMAND MODES

- User EXEC Mode
Switch>
- Privileged EXEC Mode
Switch#
- Global Configuration
Switch(config)#
- Interface Configuration
Switch(config-if)#
- Line Configuration
Switch(config-line)#

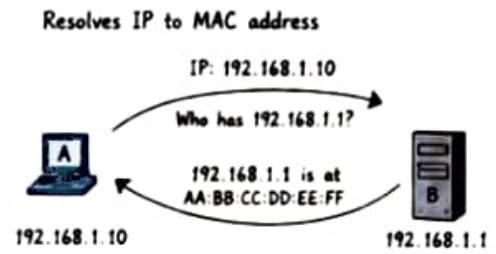
⑨ BASIC IOS COMMANDS

- enable - Enter privileged mode
- configure terminal - Enter config mode
- hostname <name> - Set hostname
- interface <type> <num> - Enter interface
- ip address <ip> <mask> - Set IP
- no shutdown - Enable interface
- show ip interface brief - Show IPs
- show running-config - Show config
- copy run start - Save config

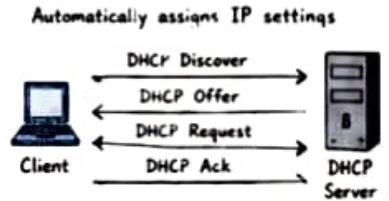
⑩ IP ADDRESS TYPES

- Unicast - One to one
 - Broadcast - One to all
(e.g., 255.255.255.255)
 - Multicast - One to many
 - Anycast - One to nearest
- 

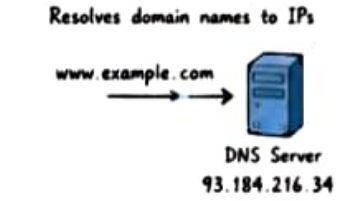
⑪ ARP (ADDRESS RESOLUTION PROTOCOL)



⑫ DHCP (DYNAMIC HOST CONFIGURATION PROTOCOL)



⑬ DNS (DOMAIN NAME SYSTEM)



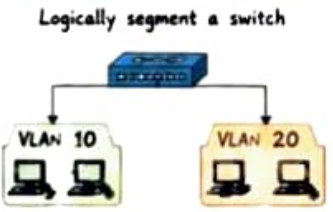
⑭ NAT (NETWORK ADDRESS TRANSLATION)



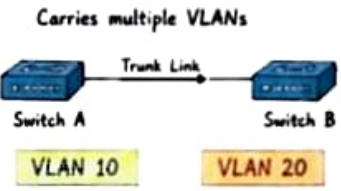
⑮ SWITCHING CONCEPTS

- MAC Address Table
- Frame Forwarding
- Flooding (Unknown Unicast)
- Broadcast Domain (per VLAN)
- Spanning Tree Protocol (STP)

⑯ VLAN (VIRTUAL LAN)



⑰ TRUNKING (802.1Q)



⑱ STP STATES

- Blocking - Blocks traffic
- Listening - Learning BPDU
- Learning - Building MAC table
- Forwarding - Forwards traffic

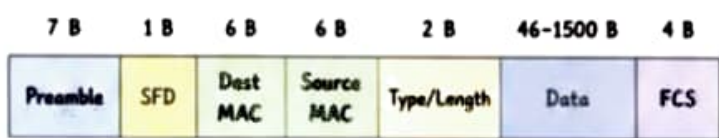
QUICK NOTES

- Always save configs
- Document your network
- Understand before you configure
- Practice show commands

- ✓ Practice labs daily
- ✓ Focus on understanding
- ✓ Troubleshoot methodically
- ✓ Think like the network

Keep Learning,
Keep Growing!

① ETHERNET FRAME FORMAT



- Preamble - Synchronization
- SFD - Start Frame Delimiter (10101011)
- Dest MAC - Destination address
- Source MAC - Source address
- Type/Length - Protocol type or length of data
- Data - Payload
- FCS - Frame Check Sequence (CRC)

② MAC ADDRESS

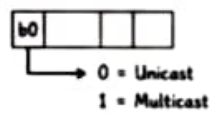
Format: 48 bits (6 bytes)

00 : 1A : 2B : 3C : 4D : 5E

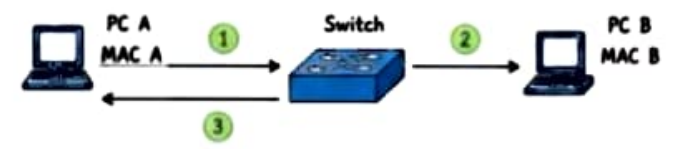
- First 24 bits - OUI (Vendor)
- Last 24 bits - Unique to device
- Written in hexadecimal
- Unicast - 0 in least significant bit
- Multicast - 1 in least significant bit
- Broadcast - FF:FF:FF:FF:FF:FF

Examples:

Unicast: 00:1A:2B:3C:4D:5E
Multicast: 01:00:5E:00:00:01
Broadcast: FF:FF:FF:FF:FF:FF



③ SWITCHING PROCESS



- 1 PC A sends frame to PC B → Switch learns MAC A on incoming port.
- 2 If MAC B is known → Switch forwards frame to that port.
If MAC B is unknown → Switch floods frame out all ports (except incoming).
- 3 PC B replies → Switch learns MAC B and forwards frame to MAC A.

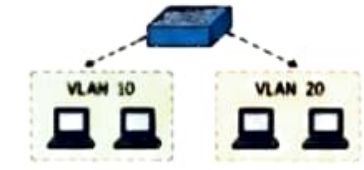
④ SWITCH MAC ADDRESS TABLE

MAC Address	VLAN	Port	Type	Aging Time
00:11:22:33:44:55	10	Fa0/1	Dynamic	120
00:11:22:33:44:66	10	Fa0/2	Dynamic	95
00:11:22:33:44:77	10	Fa0/3	Dynamic	85
00:11:22:33:44:88	20	Fa0/5	Dynamic	110
00:11:22:33:44:99	20	Fa0/6	Static	-

- Dynamic entries age out (default 300 sec).
- Static entries do not age out.
- Use: `show mac address-table`

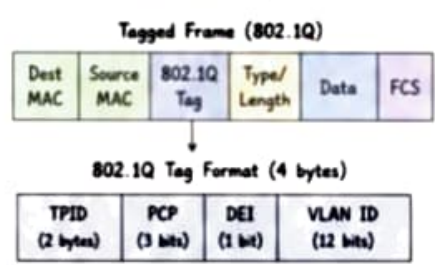
⑤ VLAN CONCEPTS

- VLAN (Virtual LAN) - Logical segmentation of a switch.
- Devices in same VLAN can communicate without router.
- Inter-VLAN communication requires a Layer 3 device (Router or L3 Switch).
- VLAN ID Range: 1 - 4094
- Default VLAN: 1
- Native VLAN (for trunks): usually 1



⑥ TRUNKING (802.1Q)

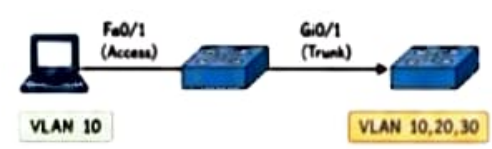
- Trunk link carries multiple VLANs.
- 802.1Q adds a 4-byte tag to Ethernet frame.



- TPID = 0x8100
- VLAN ID = 1 - 4094

⑦ ACCESS vs TRUNK PORT

Feature	Access Port	Trunk Port
Carries	Single VLAN	Multiple VLANs
Tagging	Untagged	Tagged (802.1Q)
PVID	Yes	Yes (Native VLAN)
Use	End device connection	Switch to Switch / Router connection
Default	Access	Trunk

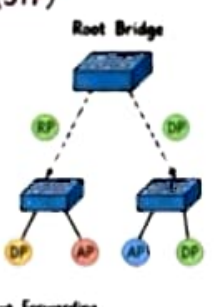


⑧ SPANNING TREE PROTOCOL (STP)

- Prevents Layer 2 loops.
- Creates a loop-free logical topology.
- Elects a Root Bridge.
- Ports have roles:

- Root Port - Best path to Root Bridge
- Designated Port - Best path on the segment
- Alternate Port - Backup path (blocking)
- Backup Port - Backup for another segment

• States: Blocking → Listening → Learning → Forwarding



⑨ COMMON SWITCH SHOW COMMANDS

- `show vlan brief` - Show VLANs
- `show interfaces status` - Port status
- `show mac address-table` - MAC table
- `show interfaces trunk` - Trunk details
- `show spanning-tree` - STP status
- `show cdp neighbors` - CDP neighbors
- `show running-config` - Running config
- `show startup-config` - Startup config

⑩ LAYER 2 SWITCH FEATURES

- MAC Address Learning & Forwarding
- VLANs
- Trunking (802.1Q)
- Spanning Tree Protocol (STP)
- Port Security
- EtherChannel (Link Aggregation)
- Storm Control
- DHCP Snooping
- Dynamic ARP Inspection (DAI)
- Private VLAN
- QoS (Trust / CoS)



QUICK NOTES

- ✓ Know OSI & TCP/IP models.
- ✓ Practice show commands.
- ✓ Document your network.
- ✓ Understand switching & VLANs.
- ✓ Labs > Theory.
- ✓ Keep it simple and secure.

Keep Learning,
Keep Growing!

① IP ROUTING BASICS

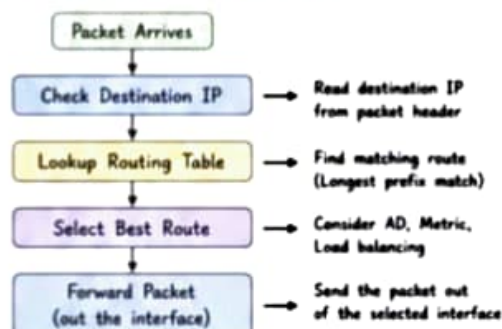
- Routers forward packets between different networks.
- Routing table is used to decide the best path.
- Routing decisions are based on destination IP address.
- Each route has an administrative distance (AD) and metric.

Routing Table Example

Destination	Mask	Next Hop	Interface	AD/Metric
0.0.0.0	0.0.0.0	192.168.1.1	G0/0	1
10.1.1.0	255.255.255.0	0.0.0.0	G0/1	0
192.168.1.0	255.255.255.0	0.0.0.0	G0/0	0

0.0.0.0/0 is the default route.
0.0.0.0 in Next Hop means "directly connected".

② ROUTER FORWARDING PROCESS



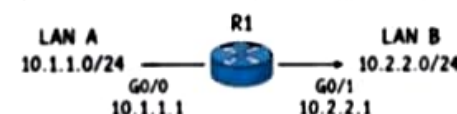
③ STATIC ROUTES

- Manually configured by network admin.
- Good for small networks.
- Not adaptive to topology changes.

`ip route <destination> <mask> <next-hop-ip> [AD]`

Examples:

`ip route 10.2.2.0 255.255.255.0 192.168.1.2`
`ip route 0.0.0.0 0.0.0.0 192.168.1.1` (Default Route)
`ip route 10.3.3.0 255.255.255.0 G0/1` (Exit Interface)

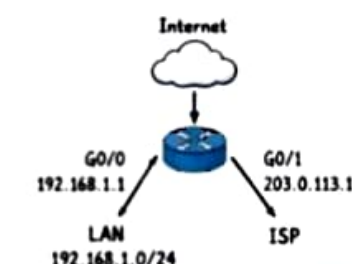


On R1 to reach LAN B:

`ip route 10.2.2.0 255.255.255.0 10.2.2.2`

④ DEFAULT ROUTE

- Used when no specific route is found for destination.
- 0.0.0.0/0 matches all destinations.
- Commonly used to reach the Internet.



R1 default route:

`ip route 0.0.0.0 0.0.0.0 203.0.113.2`

⑤ ADMINISTRATIVE DISTANCE (AD)

- AD is a trust value for the route source.
- Lower AD is preferred.
- Range: 0 (trusted) to 255 (untrusted)

Route Source	AD (Default)
Connected	0
Static	1
EIGRP (Internal)	90
OSPF	110
IS-IS	115
RIP	120
EIGRP (External)	170
BGP	20
Unknown	255

If AD is same, metric is used to choose best route.

⑥ METRIC

- Metric is a value used to determine the best path.
- Different routing protocols use different metrics.
- Lower metric is better.

Protocol	Metric
RIP	Hop Count
OSPF	Cost (Bandwidth based)
EIGRP	Composite (Bandwidth, Delay, etc.)
IS-IS	Cost
BGP	Path attributes (not a metric)

⑦ LONGEST PREFIX MATCH

- Router matches the most specific route (instead of most general).
- More 1s in the mask = more specific.

Destination/Mask	Match?
10.1.1.0/24	Match
10.1.0.0/16	Match
0.0.0.0/0	Match

← Best Match

Destination IP: 10.1.1.25

Best match: 10.1.1.0/24 (most specific)

⑧ ROUTE TYPES

C	Connected	Directly connected networks
S	Static	Manually configured routes
R	RIP	Learned via RIP
O	OSPF	Learned via OSPF
D	EIGRP	Learned via EIGRP
B	BGP	Learned via BGP
*	Candidate Default	Default route candidate
>	Best Route	Installed in routing table

Use: `show ip route`

⑨ SHOW IP ROUTE OUTPUT EXAMPLE

R1# `show ip route`

Codes: C - connected, S - static, R - RIP, O - OSPF, D - EIGRP, B - BGP, * - candidate default, > - best
Gateway of last resort is 192.168.1.1 to network 0.0.0.0

```

C    192.168.1.0/24 is directly connected, G0/0
C    10.1.1.0/24 is directly connected, G0/1
S*   0.0.0.0/0 [1/0] via 192.168.1.1
R    172.16.0.0/16 [120/1] via 192.168.1.2, 00:00:12
O    192.168.2.0/24 [110/2] via 10.1.1.2, 00:01:30
    
```

⑩ PACKET FLOW (END TO END)



- Host sends packet to default gateway.
- Router routes packet to next hop (based on routing table).
- Packet reaches destination network and is delivered.

⑪ USEFUL COMMANDS

<code>show ip interface brief</code>	- Interface status
<code>show ip route</code>	- Routing table
<code>show ip protocols</code>	- Routing protocols info
<code>show ip interface g0/0</code>	- Interface details
<code>ping <ip></code>	- Test connectivity
<code>traceroute <ip></code>	- Trace the path
<code>debug ip routing</code>	- Debug routing

QUICK NOTES

- ✓ Understand routing concepts deeply.
- ✓ Practice routing configurations.
- ✓ Use show commands to verify.
- ✓ Know AD and Metrics.
- ✓ Longest prefix match is key.
- ✓ Always save configuration.

Keep Learning,
Keep Growing!

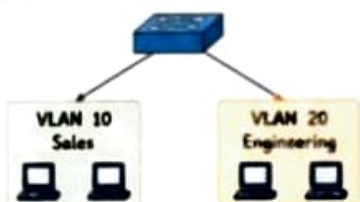
① SWITCHING FUNDAMENTALS

- Switches operate at Layer 2 (Data Link).
- Use MAC addresses to forward frames.
- Each port is a separate collision domain.
- Uses CAM table (Content Addressable Memory).
- Unknown unicast, broadcast and multicast flood out all ports in the VLAN.



④ VLAN (VIRTUAL LAN)

- Logical segmentation of a switch.
- Devices in same VLAN can communicate.
- Different VLANs need a Layer 3 device (router or L3 switch) to communicate.



- Commands:

```
vlan 10
name Sales
interface range fa0/1 - 2
switchport mode access
switchport access vlan 10
```

⑦ SPANNING TREE PROTOCOL (STP)

- Prevents Layer 2 loops.
- Creates a loop-free logical topology.
- Elects a Root Bridge.
- Ports have roles:

- Root Port (RP)** - Best path to Root Bridge
- Designated Port (DP)** - Best path on segment
- Alternate Port (AP)** - Backup (blocking)
- Backup Port (BP)** - Backup for other segment

- States: Blocking → Listening → Learning → Forwarding



⑧ PORT STATES (STP)

State	Description
Blocking	Does not forward frames
Listening	Listens for BPDUs
Learning	Learns MAC addresses
Forwarding	Forwards frames

- Command: `show spanning-tree`

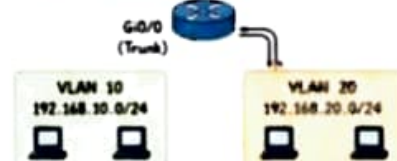
⑩ INTER-VLAN ROUTING

- Devices in different VLANs need Layer 3 to communicate.

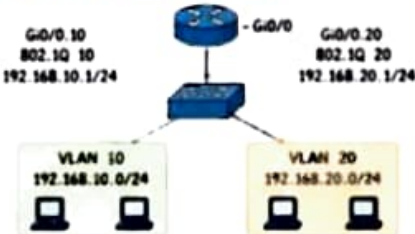
- Methods:

 - Router-on-a-Stick (ROAS)
 - Layer 3 Switch (SVI)
 - Separate Physical Router

1. Router-on-a-Stick



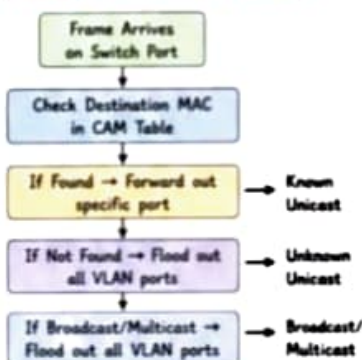
⑪ SUBINTERFACES (ROAS)



- Commands (Router):

```
interface g0/0.10
encapsulation dot1q 10
ip address 192.168.10.1 255.255.255.0
interface g0/0.20
encapsulation dot1q 20
ip address 192.168.20.1 255.255.255.0
```

③ FRAME FORWARDING PROCESS



⑥ TRUNK MODES

Mode	Description
access	Port belongs to one VLAN (Untagged frames only)
trunk	Carries multiple VLANs (Tagged frames with 802.1Q)
dynamic auto	Negotiates trunk with DTP (Forms trunk if peer is desirable)
dynamic desirable	Actively tries to form trunk with DTP
nonegotiate	Disables DTP on the port

- Command:

```
interface gi0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30
```

⑨ STP MODES

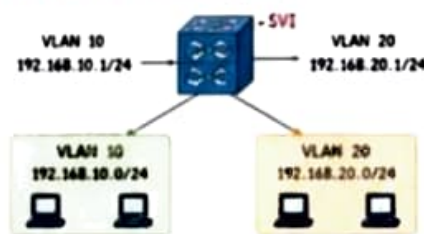
Mode	Description
PVST+	Cisco proprietary (default)
Rapid-PVST+	Faster convergence
MSTP	Multiple Spanning Tree Protocol (802.1s)

- Commands:

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10 priority 4096
```

Tip: Lower bridge priority wins.
Range: 0 - 61440 (default: 32768)

⑫ LAYER 3 SWITCH (SVI)



- Commands:

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
interface vlan 20
ip address 192.168.20.1 255.255.255.0
ip routing
```

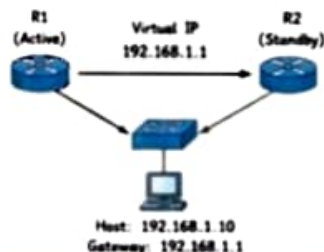
QUICK NOTES

- ✓ Understand switching & VLANs deeply.
- ✓ Practice STP and Trunking.
- ✓ Know inter-VLAN routing methods.
- ✓ Use show commands to verify.
- ✓ Document your network.
- ✓ Practice, Practice, Practice!

Keep Learning,
Keep Growing!

① FIRST HOP REDUNDANCY

- Provides backup default gateway.
- Router redundancy for better availability.
- Protocols: HSRP (Cisco), VRRP, GLBP.



```

HSRP Example (Router)
R(config)# interface g0/0
R(config-if)# standby 1 ip 192.168.1.1
R(config-if)# standby 1 priority 110
R(config-if)# standby 1 preempt
    
```

④ ACL WILDCARD MASKS

- Wildcard mask is the inverse of subnet mask.
- 0 = Must match 1 = Ignore

Subnet Mask	Wildcard Mask	Meaning
255.255.255.255	0.0.0.0	Single host
255.255.255.0	0.0.0.255	/24 network
255.255.0.0	0.0.255.255	/16 network
255.0.0.0	0.255.255.255	/8 network
0.0.0.0	255.255.255.255	Any

Example

Permit 192.168.1.0/24

```
access-list 1 permit 192.168.1.0 0.0.0.255
```

⑦ STATIC NAT EXAMPLE

- Maps 192.168.1.10 to 203.0.113.10



```

R(config)# ip nat inside source static 192.168.1.10 203.0.113.10
R(config)# interface g0/0
R(config-if)# ip nat inside
R(config)# interface g0/1
R(config-if)# ip nat outside
    
```

Inside Local	Inside Global	Outside Local	Outside Global
192.168.1.10	203.0.113.10	203.0.113.10	203.0.113.10

⑩ IP SLA (SERVICE LEVEL AGREEMENT)

- Measures network performance.
- Uses probes (ICMP, TCP, UDP, DNS).
- Results can trigger actions.
- Configure and schedule operations.

```

R(config)# ip sla 1
R(config-ip-sla)# icmp-echo 8.8.8.8 source-interface g0/0
R(config-ip-sla)# frequency 30
R(config)# ip sla schedule 1 life forever start-time now
R# show ip sla statistics 1
    
```

② HSRP STATES

State	Description
Active	Forwards packets for the group.
Standby	Backup router, takes over if Active fails.
Listen	Initializing, not yet participating.
Speak	Exchanging hellos, not yet elected.
Init	No hello packets received.

HSRP Timers (Defaults)

- Hello Time: 3 seconds
- Hold Time: 10 seconds

HSRP Priority (Default 100)

- Higher priority = Active router.
- Range: 0 - 255 (255 = highest)

⑤ EXTENDED ACL EXAMPLE

Goal: Allow HTTP (80) from 192.168.1.0/24 to any, deny the rest.

```

access-list 101 permit tcp
192.168.1.0 0.0.0.255 any eq 80
access-list 101 deny ip any any
    
```

Applied Inbound on G0/0

```

Interface g0/0
ip access-group 101 in
    
```



⑧ PAT (NAT OVERLOAD) EXAMPLE

- Many inside hosts share one public IP.
- Uses port numbers to distinguish sessions.



```

R(config)# access-list 1 permit 192.168.1.0 0.0.0.255
R(config)# ip nat inside source list 1 interface g0/1 overload
R(config)# interface g0/0
ip nat inside
R(config)# interface g0/1
ip nat outside
    
```

③ ACCESS CONTROL LISTS (ACL)

- Used to filter traffic.
- Applied inbound or outbound.
- Types: Standard, Extended, Named.

Standard ACL	Extended ACL
- Matches source IP only. - ACL range: 1 - 99, 1300 - 1999.	- Matches source, dest, protocol, port, etc. - ACL range: 100 - 199, 2000 - 2699.

Example: Block host 192.168.1.10

```

R(config)# access-list 1 deny host 192.168.1.10
R(config)# access-list 1 permit any
R(config)# interface g0/0
R(config-if)# ip access-group 1 in
    
```

⑥ NAT OVERVIEW

- Translates private IP addresses to public IP.
- Types: Static NAT, Dynamic NAT, PAT (NAT Overload).
- Configured on the border router.



NAT Type	Description
Static NAT	1:1 mapping (inside local ↔ inside global)
Dynamic NAT	Many-Many (uses pool of public IPs)
PAT (NAT Overload)	Many:1 (uses one public IP + ports)

⑨ NAT TERMS

Term	Meaning
Inside Local	Private IP of internal host
Inside Global	Public IP used to represent inside local
Outside Local	Public IP of external host
Outside Global	Actual public IP of external host



⑪ SYSLOG & SNMP

Syslog

- Logs events and errors.
- Sent to syslog server.
- Levels: 0 (emerg) - 7 (debug).

```

R(config)# logging host 192.168.1.100
R(config)# logging trap warnings
    
```

SNMP

- Network monitoring protocol.
- Uses community strings (v2c).

```

R(config)# snmp-server community public RO
R(config)# snmp-server location MainOffice
R(config)# snmp-server contact Admin
    
```

⑫ TIME SYNCHRONIZATION (NTP)

- Keeps device clocks accurate.
- Uses NTP servers.

```

R(config)# ntp server 0.pool.ntp.org
R(config)# ntp server 1.pool.ntp.org
R(config)# clock timezone PST -8
R(config)# clock summer-time PDT recurring
R# show clock
    
```

COMMON PORT REMINDERS

- 20/21 - FTP
- 22 - SSH
- 23 - Telnet
- 25 - SMTP
- 53 - DNS
- 80 - HTTP
- 110 - POP3
- 143 - IMAP
- 161/162 - SNMP

QUICK NOTES

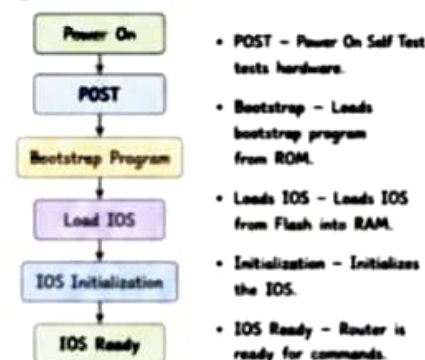
- ✓ Save configurations regularly.
- ✓ Understand before you configure.

- ✓ Practice labs and real scenarios.
- ✓ Document and review often.

- ✓ Use show commands to verify.
- ✓ Think like the network!

Keep Learning,
Keep Growing!

① ROUTER BOOT PROCESS



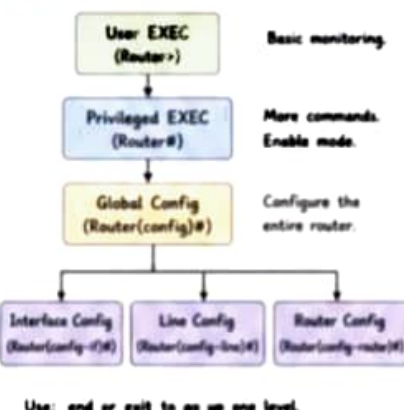
② ROUTER FILE SYSTEMS

File System	Description
Flash:	Non-volatile memory (stores IOS)
NVRAM:	Stores startup configuration
RAM:	Volatile memory (running config, routing tables, ARP cache)
ROM:	Read-only memory (bootstrap)
TFTP Server:	Remote server (backup, IOS transfer)
USB:	External storage (on some routers)

③ SHOW COMMANDS (ROUTER)

show ip interface brief	Brief status of all interfaces
show ip route	Routing table
show running-config	Current running configuration
show startup-config	Startup configuration
show version	IOS version info
show interfaces	Detailed interface status
show protocols	Routing protocols status
show cdp neighbors	CDP neighbors
show arp	ARP table
show ip ospf neighbor	OSPF neighbors
show ip eigrp neighbors	EIGRP neighbors

④ ROUTER MODES



⑤ IP ADDRESS CLASSES (CLASSFUL)

Class	First Octet Range	Default Mask	Max Hosts
A	1 - 126	255.0.0.0	16,777,214
B	128 - 191	255.255.0.0	65,534
C	192 - 223	255.255.255.0	254
D	224 - 239	Multicast	N/A
E	240 - 255	Experimental	N/A

Private Address Ranges (Not routed on Internet)

- Class A: 10.0.0.0 - 10.255.255.255 (/8)
- Class B: 172.16.0.0 - 172.31.255.255 (/16)
- Class C: 192.168.0.0 - 192.168.255.255 (/24)

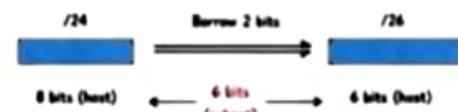
⑥ SUBNETTING EXAMPLE

Network: 192.168.1.0/24
Need 4 subnets

Step 1: Borrow 2 bits
 $2^2 = 4$ subnets

New Prefix: /26
Mask: 255.255.255.192

Subnet	Network ID	Host Range	Broadcast
1	192.168.1.0	1 - 62	63
2	192.168.1.64	65 - 126	127
3	192.168.1.128	129 - 190	191
4	192.168.1.192	193 - 254	255



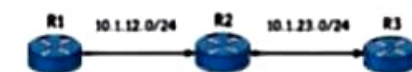
⑦ ROUTING PROTOCOL OVERVIEW

Protocol	Type	Class	Metric	Admin Distance
RIP	Distance Vector	IGP	Hop Count	120
OSPF	Link State	IGP	Cost	110
EIGRP	Advanced DV	IGP	Composite	90
IS-IS	Link State	IGP	Cost	115
BGP	Path Vector	EGP	Path Attributes	20

Lower Administrative Distance (AD) is preferred.

⑧ RIP (ROUTING INFORMATION PROTOCOL)

- Distance Vector Protocol.
- Uses Hop Count as metric.
- Max 15 hops (16 = unreachable).
- Updates every 30 seconds.
- Uses UDP port 520.

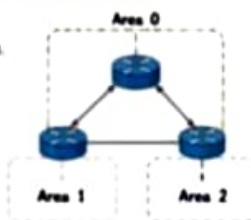


```

R(config-router)# router rip
R(config-router)# network 10.0.0.0
(Enables RIP on interfaces of that network)
  
```

⑨ OSPF (OPEN SHORTEST PATH FIRST)

- Link State Protocol.
- Uses Cost (bandwidth based).
- Supports VLSM, CIDR.
- Faster convergence.
- Uses IP protocol 89.



```

R(config-router)# router ospf 1
R(config-router)# network 19.1.1.0 0.0.0.255 area 0
(Process ID is locally significant)
  
```

⑩ EIGRP (ENHANCED IGRP)

- Advanced Distance Vector.
- Uses Composite metric: Bandwidth, Delay, Reliability, Load, MTU.
- Fast convergence.
- Uses IP protocol 88.
- Supports VLSM, CIDR.



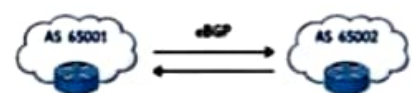
```

R(config-router)# router eigrp 100
R(config-router)# network 10.0.0.0
R(config-router)# no auto-summary
  
```

Tip: Always disable auto-summary.

⑪ BGP (BORDER GATEWAY PROTOCOL)

- Path Vector Protocol (EGP).
- Uses TCP port 179.
- Runs between AS (Autonomous Systems).
- Scalability & Policy based routing.



```

R(config)# router bgp 65001
R(config-router)# neighbor 192.168.1.2 remote-as 65002
  
```

Tip: Use show ip bgp to see BGP table.

⑫ DEFAULT ROUTE

- Used when no specific route is found.
- Router forwards packet to default gateway.
- 0.0.0.0/0 matches all destinations.



```

R(config)# ip route 0.0.0.0 0.0.0.0 10.2.2.1
  
```



QUICK NOTES

- ✓ Understand concepts before commands.
- ✓ Practice labs on real or simulated devices.
- ✓ Use show commands to verify.

- ✓ Know your OSI, TCP/IP, and protocols.
- ✓ Document your network.
- ✓ Troubleshooting = Observe, Analyze, Act.

Keep Learning,
Keep Growing!

① OSPF AREA TYPES

- OSPF uses areas to scale.
- Area 0 is the Backbone Area.
- Other areas connect to Area 0.
- Types:

Area Type	Description
Backbone	Area 0. All other areas must connect to it.
Standard	Normal area. Contains routers and networks.
Stub	Does not advertise external routes (Type 3/5 LSAs).
Totally Stubby	Does not advertise Type 3, Type 4, or Type 5 LSAs.
NSSA	Allows Type 7 LSAs (converted to Type 5 at ABR).

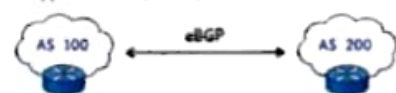
② OSPF NEIGHBOR STATES

State	Description
Down	No hello packets received.
Attempt	Trying to form adjacency.
Init	Hello packets sent, but router ID not seen.
2-Way	Router ID received.
ExStart	Master/Slave decided.
Exchange	Database description exchanged.
Loading	Link State Requests sent.
Full	Adjacency complete.

💡 Command: `show ip ospf neighbor`

③ EIGRP OVERVIEW

- Cisco proprietary Advanced Distance Vector.
- Uses DUAL algorithm.
- Fast convergence.
- Metrics: Bandwidth, Delay (default).
- Administrative Distance: 90 (internal), 170 (external).
- Supports VLSM, CIDR, Summarization.



```
R(config)# router eigrp 100
R(config-router)# network 192.168.1.0 0.0.0.255
R(config-router)# no auto-summary
R(config-router)# passive-interface default
R(config-router)# no passive-interface g0/0
```

④ EIGRP METRIC (DEFAULT)

$$\text{Metric} = [K1 \times BW + K2 \times BW / (256 - \text{Load}) + K3 \times \text{Delay}] \times [K5 / (K4 + \text{Reliability})]$$

Default K values: K1=1, K2=0, K3=1, K4=0, K5=0, Reliability=255, Load=1

Parameter	Description	Unit
Bandwidth	Minimum bandwidth along path	Kbps
Delay	Total delay along path	10 µs
Reliability	Link reliability	(1-255)
Load	Link load	(1-255)

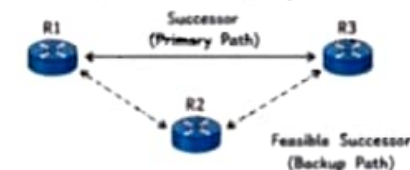
💡 Tip: By default, only Bandwidth and Delay are used in the metric.

⑤ EIGRP ROUTE TYPES

Code	Route Type	Description
D	Internal	EIGRP learned within AS
EX	External	EIGRP learned from another AS
E1	External Type 1	External with default metric
E2	External Type 2	External with metric
i	Internal Summary	Summarized within AS
>	Candidate Default	Potential default route
*	Successor	Best route
P	Feasible Successor	Backup route

⑥ EIGRP FEASIBLE SUCCESSOR

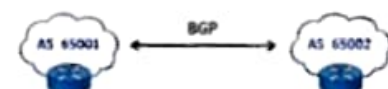
- Backup route that can be used without querying neighbors.
- Condition: Feasible Distance < Reported Distance.



```
R(config)# router eigrp 100
R(config-router)# eigrp log-neighbor-changes
R(config-router)# show ip eigrp topology
R(config-router)# show ip route eigrp
```

⑦ BGP OVERVIEW

- Exterior Gateway Protocol (Path Vector).
- Runs between Autonomous Systems.
- Uses TCP port 179.
- Administrative Distance: 20.
- Best path based on attributes (not metric).



```
R(config)# router bgp 65001
R(config-router)# neighbor 192.168.1.2 remote-as 65002
R(config-router)# network 192.168.1.0 mask 255.255.255.0
R(config-router)# show ip bgp summary
```

⑧ BGP PATH ATTRIBUTES

Attribute	Description
Weight	Cisco specific, local to router
Local Preference	Inbound traffic control
AS Path	List of ASes the route traversed
Origin	IGP, EGP, or Incomplete
MED	Multi-Exit Discriminator
Next Hop	IP of next hop
Community	Used for tagging (policy)
Originator ID	Used in route reflection
Cluster List	Used in route reflection

⑨ BGP ROUTE SELECTION ORDER

- Highest Weight (Cisco)
- Highest Local Preference
- Locally originated route
- Shortest AS Path
- Lowest Origin (IGP < EGP < Incomplete)
- Lowest MED
- eBGP path over iBGP path
- Lowest IGP metric to Next Hop
- Oldest path
- Lowest Router ID
- Lowest Neighbor IP address

```
R(config-router)# show ip bgp
R(config-router)# show ip bgp neighbors
```

⑩ BGP NEXT-HOP

- Next-hop is the IP address of the router that will forward the packet to the destination.
- By default, iBGP does not change next-hop.
- Use next-hop-self to set next-hop as local router.



```
R(config-router)# neighbor 192.168.1.1 next-hop-self
```

⑪ ROUTE SUMMARIZATION

- Reduces routing table size.
- Advertise a summary instead of many specific routes.

Example:
Instead of advertising:
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24
192.168.4.0/24
Advertise summary:
192.168.0.0/22

```
R(config)# interface g0/0
R(config-if)# ip summary-address eigrp
100 192.168.0.0 255.255.252.0
```

⑫ DEFAULT ROUTE ORIGIN

- Inject default route into routing protocol.
- Used for Internet access.
- EIGRP Example:

```
R(config)# ip route 0.0.0.0 0.0.0.0 <next-hop>
R(config)# router eigrp 100
R(config-router)# default-information originate
```

- OSPF Example:

```
R(config)# ip route 0.0.0.0 0.0.0.0 <next-hop>
R(config)# router ospf 1
R(config-router)# default-information originate
```

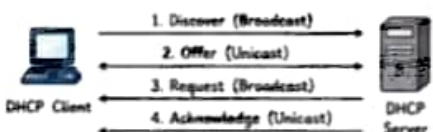
QUICK NOTES

- Understand routing protocols deeply.
- Know metrics, AD, and path selection.
- Practice configurations and verification.
- Use show commands to monitor.
- Troubleshoot step by step.
- Lab, Practice, Repeat!

Keep Learning,
Keep Growing!

① DHCP OVERVIEW

- Dynamic Host Configuration Protocol.
- Automatically assigns IP configuration.
- Uses UDP ports 67 (server) and 68 (client).
- DHCP process (DORA):
Discover → Offer → Request → Acknowledge
(D) (O) (R) (A)



```
R(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.20
R(config)# ip dhcp pool LAN1
R(dhcp-config)# network 192.168.1.0 255.255.255.0
R(dhcp-config)# default-router 192.168.1.1
R(dhcp-config)# dns-server 8.8.8.8
```

② DHCP MESSAGE TYPES

Message	Description
Discover	Client broadcasts to find DHCP server.
Offer	Server offers an IP address to the client.
Request	Client requests the offered IP address.
Acknowledge	Server acknowledges and leases the IP address.
NAK	Server denies the request.
Release	Client releases the IP address.
Decline	Client declines the offered IP.
Inform	Client requests configuration information only.

Tip: DHCP lease information
show ip dhcp binding

③ IPV6 OVERVIEW

- 128-bit address (e.g., 2001:db8:1::1/64)
- Simplified header for faster processing.
- Supports more addresses (2^{128}).
- Address types:

Address Type	Example
Global Unicast	2001:db8:1::1
Link-Local	FE80::1
Unique Local (ULA)	FDD0::1
Multicast	FF02::1
Anycast	2001:db8::1

```
R(config)# ipv6 unicast-routing
R(config)# interface g0/0
R(config-if)# ipv6 address 2001:db8:1::1/64
R(config-if)# no shutdown
Show: show ipv6 interface brief
```

④ IPV6 ADDRESS TYPES

Type	Description
Global Unicast	Routeable on the IPv6 Internet.
Link-Local	Used on local link only (FE80::/10).
Unique Local (ULA)	Private addressing (FDD0::/7).
Multicast	One-to-many communication.
Anycast	One-to-nearest communication.
Loopback	::1 (used by device itself).
Unspecified	:: (used before address is assigned).

★ IPv6 Link-Local address is automatically configured on every interface.

⑤ IPV6 NEIGHBOR DISCOVERY (ND)

- Replaces ARP, ICMP redirects, and more.
- Uses ICMPv6 messages.

ND Message	Purpose
Router Solicitation (RS)	Host asks for routers.
Router Advertisement (RA)	Router advertises info.
Neighbor Solicitation (NS)	Finds MAC address.
Neighbor Advertisement (NA)	Reply to NS message.
Redirect	Tells host a better router.
Duplicate Address Detection (DAD)	Checks for duplicate IPv6 address.

```
R(config)# ipv6 nd ra suppress
R(config)# ipv6 nd prefix 2001:db8:1::/64 86400
R(config)# show ipv6 neighbors
```

⑥ IPV6 SLAAC & DHCPV6

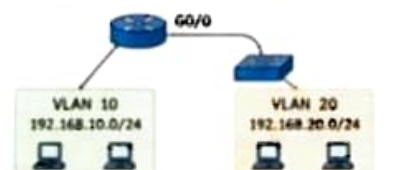
- SLAAC (Stateless):
- Uses RA messages to get prefix.
- Host builds address using EUI-64 or random interface ID.

- DHCPv6 (Stateful):
- Provides other info (DNS, domain, etc).
- DHCPv6 uses UDP ports 546 (client) and 547 (server).

```
R(config)# ipv6 unicast-routing
R(config)# ipv6 dhcp pool IPV6-POOL
R(dhcpv6-config)# dns-server 2001:4860:4860::8888
R(config)# interface g0/0
R(config-if)# ipv6 nd managed-config-flag
R(config-if)# ipv6 nd other-config-flag
R(config-if)# no shutdown
```

⑦ INTER-VLAN ROUTING (ROUTER-ON-A-STICK)

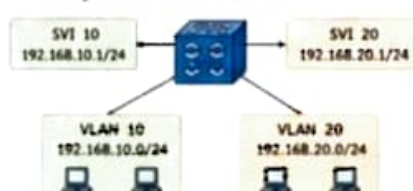
- Single physical link between router and switch.
- Router subinterfaces for each VLAN.



```
R(config)# interface g0/0:10
R(config-subif)# encapsulation dot1q 10
R(config-subif)# ip address 192.168.10.1 255.255.255.0
R(config)# interface g0/0:20
R(config-subif)# encapsulation dot1q 20
R(config-subif)# ip address 192.168.20.1 255.255.255.0
R(config-subif)# no shutdown
```

⑧ INTER-VLAN ROUTING (LAYER 3 SWITCH)

- Layer 3 switch with SVIs (Switch Virtual Interfaces).
- Routing done in switch hardware.



```
SW(config)# ip routing
SW(config)# interface vlan 10
SW(config-if)# ip address 192.168.10.1 255.255.255.0
SW(config)# interface vlan 20
SW(config-if)# ip address 192.168.20.1 255.255.255.0
SW(config-if)# no shutdown
```

⑨ ROUTE SUMMARIZATION

- Reduces size of routing table.
- Advertise a single summary route.
- Example:

Block Size	/Prefix
1	/32
2	/31
4	/30
8	/29
16	/28
...	...
4096	/20

```
R(config)# ip route
192.168.0.0 255.255.240.0
<next-hop>
```

Verify:
show ip route
show ip protocols

⑩ DISTANCE ADMINISTRATIVE PREFERENCE

- Lower AD is preferred.
- Ties broken by metric.

Route Source	Administrative Distance
Connected	0
Static	1
IGRP (Internal)	90
OSPF	110
IS-IS	115
RIP	120
IGRP (External)	170
BGP	20 (eBGP), 200 (iBGP)

Preferred (Lower AD) ↑
↓ Less Preferred (Higher AD)

⑪ ROUTING TABLE CODES

Code	Meaning
C	Connected
S	Static
R	RIP
O	OSPF
D	EIGRP
B	BGP
L	Local
*	Candidate Default
>	Best Route

Example: show ip route
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
S 0.0.0.0/0 [1/0] via 192.168.1.1
O 10.1.1.0/24 [110/2] via 10.1.1.2
D 172.16.0.0/16 [90/4] via 10.1.1.3
B* 0.0.0.0/0 [20/0] via 192.168.2.1

⑫ USEFUL SHOW COMMANDS

Command	Purpose
show ip interface brief	IP interface summary
show ip route	Routing table
show ip protocols	Routing protocols status
show ip ospf neighbor	OSPF neighbors
show ip eigrp neighbors	EIGRP neighbors
show running-config	Current configuration
show startup-config	Saved configuration
show version	System information
show cdp neighbors	CDP neighbors
show vlan brief	VLAN summary

QUICK NOTES

- ✓ Understand IPv4 & IPv6 deeply.
- ✓ Practice subnetting and summarization.
- ✓ Know routing protocols and admin distance.

- ✓ Use 'show' commands to verify everything.
- ✓ Document and save your configurations.
- ✓ Practice labs and real-world scenarios.
- ✓ Troubleshoot methodically.
- ✓ Stay consistent and keep learning.

Keep Learning.
Keep Growing!

① STP PORT ROLES

- Root Port (RP):** Best path from non-root switch to Root Bridge.
- Designated Port (DP):** Best path from segment to Root Bridge.
- Alternate Port (AP):** Backup path (blocking state).
- Backup Port (BP):** Backup for Root Port.

② STP STATES

State	Description
Blocking	Does not forward frames. Listens for BPDUs.
Listening	Listens for BPDUs. Builds MAC table.
Learning	Builds MAC table. Does not forward frames.
Forwarding	Forwards frames.

Default timers (802.1D):

- Hello Time: 2 sec
- Max Age: 20 sec
- Forward Delay: 15 sec

```
show spanning-tree
show spanning-tree vlan <vlan-id>
```

③ ETHERCHANNEL (LINK AGGREGATION)

- Bundles multiple physical links into one logical link.
- Increases bandwidth and provides redundancy.
- Protocols: LACP (802.3ad) or PAgP (Cisco).

Benefits

- Higher bandwidth
- Link redundancy
- Load balancing

```
SW1(config)# interface range g0/1 - 2
SW1(config-if-range)# channel-group 1 mode active ! LACP
SW1(config)# interface port-channel 1
SW1(config-if)# switchport mode trunk
```

④ VLAN TRUNKING PROTOCOL (VTP)

- Synchronizes VLANs across switches in a domain.
- Modes: Server, Client, Transparent.
- Uses UDP port 269.
- Versions: 1, 2, 3 (version 2 is common).

Mode	Description
Server	Creates, edits, and deletes VLANs. Saves VLAN info to NVRAM.
Client	Cannot create/edit VLANs. Learns VLANs from server.
Transparent	Does not participate in VLAN updates. Forwards VTP messages.

```
SW1(config)# vtp domain CCNA
SW1(config)# vtp mode server
SW1(config)# vtp password cisco
SW1(config)# show vtp status
```

⑤ VLAN PRUNING

- Removes unnecessary VLAN traffic on trunks.
- Saves bandwidth.
- Works with VTP (requires server mode).

```
SW1(config)# vtp mode server
SW1(config)# vtp pruning
SW1(config)# interface g0/1
SW1(config-if)# switchport trunk pruning vlan 10,20
```

⑥ FIRST-HOP REDUNDANCY PROTOCOLS

HSRP (Hot Standby Router Protocol)

- Virtual IP and MAC address.
- Active and Standby routers.
- Uses UDP 1985.

VRRP (Virtual Router Redundancy Protocol)

- Open standard (RFC 5798).
- Uses IP protocol 112.
- Similar to HSRP.

⑦ QoS (QUALITY OF SERVICE)

- Prioritizes network traffic.
- Components:
 1. Classification (Identify traffic)
 2. Marking (Set priority)
 3. Queuing & Scheduling (Manage traffic)
 4. Policing/Shaping (Control rate)

Traffic Class	DSCP Value (Example)
Voice	EF (Expedited Forwarding) - 46
Video	AF41 - 34
Critical Data	AF31 - 26
Best Effort	Default - 0

```
R(config)# class-map match-any VOICE
R(config-cmap)# match dscp ef
R(config)# policy-map QOS-POLICY
R(config-pmap)# class VOICE
R(config-pmap-c)# priority percent 30
```

⑧ SNMP OVERVIEW

- Simple Network Management Protocol.
- Used for monitoring and management.
- Uses UDP 161 (queries) and 162 (traps).
- Versions: v1, v2c, v3.

Component	Description
Manager (NMS)	Sends requests and receives responses.
Agent	Runs on device, stores MIB info.
MIB	Management Information Base.
OID	Object Identifier (unique number).
PDU	Protocol Data Unit (message).

```
R(config)# snmp-server community public RO
R(config)# snmp-server location DataCenter
R(config)# snmp-server contact Admin
R(config)# snmp-server host 192.168.1.100 version 2c public
```

⑨ SYSLOG LEVELS (SEVERITY)

Level		Description
0	Emergency	System is unusable
1	Alert	Action must be taken immediately
2	Critical	Critical conditions
3	Error	Error conditions
4	Warning	Warning conditions
5	Notification	Normal but significant condition
6	Informational	Informational messages
7	Debug	Debug-level messages

```
R(config)# logging host 192.168.1.100
R(config)# logging trap warnings
R(config)# logging facility local7
```

Tip: Lower numbers = higher severity.

⑩ COMMON PORT NUMBERS

Service	Port	Service	Port
HTTP	80	DHCP Server	67
HTTPS	443	DHCP Client	68
FTP	21	SNMP	161/162
TFTP	69	Sylog	514
SSH	22	NTP	123
Telnet	23	RIP	520
SMTP	25	BGP	179
DNS	53	LDAP	389

⑪ CABLE TYPES

Cable	Use	Connector
Straight-Through	Unlike devices (Switch -> Router, PC -> Switch)	
Crossover	Like devices (Switch -> Switch, Router -> Router)	
Console (Roller)	PC -> Console port (Management)	

⑫ DEVICE PASSWORDS

Type	Description
Console Password	Secures console access.
Enable Secret	Protects privileged EXEC mode.
VTY Password	Secures remote access (Telnet/SSH)

```
R(config)# enable secret class
R(config)# line console 0
R(config-line)# password cisco
R(config-line)# login
R(config)# line vty 0 4
R(config-line)# password cisco
R(config-line)# login
R(config-line)# transport input ssh
```

① NETWORK SECURITY BASICS

- Protects confidentiality, integrity & availability.
- Key concepts: AAA, ACL, VPN, Firewall.
- Threats: Malware, Phishing, DoS/DDoS, MITM.
- Best Practices: Least privilege, updates, backups.

AAA	
Component	Purpose
Authentication	Verifies identity (Who you are)
Authorization	Determines access (What you can do)
Accounting	Logs user activity (What you did)

```

R(config)# aaa new-model
R(config)# aaa authentication login default local
R(config)# aaa authorization exec default local
R(config)# aaa accounting exec default start-stop
  
```

④ DEVICE HARDENING

- Secure device access and management.
- Disable unused services.
- Use strong passwords & encryption.
- Use SSH, not Telnet.

Feature	Recommendation
Passwords	Use strong, complex passwords
Management Access	SSH only, disable Telnet/HTTP
Unused Services	Disable (CDP, HTTP, etc.)
Banner	Display legal warning banner
Logging	Enable logging & monitoring

```

R(config)# ip domain-name cisco.local
R(config)# crypto key generate rsa modulus 2048
R(config)# ip ssh version 2
R(config)# line vty 0 4
R(config-line)# transport input ssh
R(config)# banner motd ^C Authorized Access Only! ^C
  
```

⑦ COMMON SUBNETS (QUICK REFERENCE)

Prefix	Subnet Mask	Usable Hosts
/8	255.0.0.0	16,777,214
/16	255.255.0.0	65,534
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14
/30	255.255.255.252	2
/32	255.255.255.255	1 (Host route)

★ Usable Hosts = $2^n - 2$ (except /31 and /32)

⑩ IPV6 ESSENTIALS

- 128-bit addressing.
- Example: 2001:db8:acad:1::1/64

Feature	IPv4	IPv6
Address Size	32-bit	128-bit
Notation	Dotted Decimal	Hexadecimal
Broadcast	Yes	No (Uses Multicast)
NAT	Common	Not Needed
Auto Config	Manual/DHCP	SLAAC/DHCPv6

```

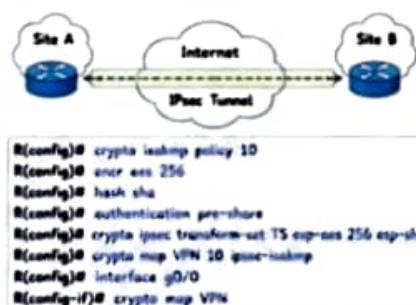
R(config)# ipv6 unicast-routing
R(config)# interface g0/0
R(config-if)# ipv6 address 2001:db8:1::1/64
R(config-if)# no shutdown
  
```

QUICK NOTES

- Understand the fundamentals deeply.
- Practice in real or simulated environments.
- Use official Cisco resources and labs.
- Review regularly and stay consistent.

② IPSEC VPN OVERVIEW

- Provides secure site-to-site or remote access.
- Uses IKE (Phase 1) and IPsec (Phase 2).
- Modes: Transport & Tunnel.
- Encrypts data and authenticates peers.



⑤ MONITORING & TROUBLESHOOTING

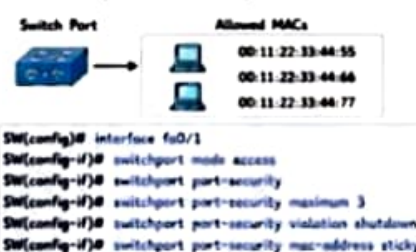
- Use show, debug, and tools to monitor network.
- Follow a systematic approach.

Troubleshooting Steps	
1. Define the problem	
2. Gather information	
3. Analyze the cause	
4. Implement a fix	
5. Verify full connectivity	
6. Document the solution	

Useful Commands	
show interface	Interface status
show ip route	Routing table
show running-config	Current config
show ip protocols	Routing protocols
show version	Device info
ping / traceroute	Connectivity test
debug ip <feature>	Detailed debugging

⑧ PORT SECURITY (SWITCHING)

- Restricts number of MAC addresses on a port.
- Protects against MAC flooding attacks.



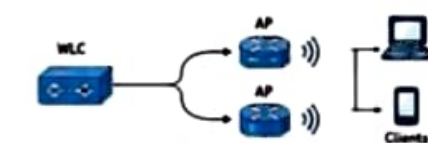
💡 Tip: Use 'sticky' to learn and save MAC addresses.

⑪ LAB TIPS

- Read requirements carefully.
- Draw the topology and plan IPs.
- Configure step by step and test.
- Verify with show commands.
- Save configuration frequently.
- Document your steps and learn.

③ WIRELESS LAN (WLAN)

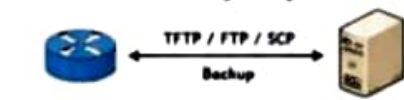
- Components: AP, WLC, SSID, Clients.
- Modes: Lightweight (CAPWAP) & Autonomous.
- Security: WPA2/WPA3, 802.1X recommended.



Security	Description
Open	No encryption (Not secure)
WEP	Legacy, weak encryption
WPA2-PSK	Pre-shared key (Good)
WPA2/3-Enterprise	Uses 802.1X & RADIUS (Best)

⑥ BACKUP & CONFIGURATION MANAGEMENT

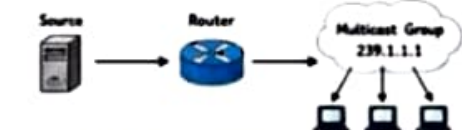
- Always backup running configuration.
- Save to TFTP/FTP/USB/SCP server.
- Version control and change management.



💡 Tip: Save often and keep backups safe!

⑨ MULTICAST BASICS

- Efficient one-to-many communication.
- Uses IGMP (hosts) and PIM (routers).



Common Multicast Addresses	
224.0.0.0 - 224.0.0.255	Link-Local (Not routed)
239.0.0.0 - 239.255.255.255	Admin Scoped

⑫ EXAM TIPS

- Understand concepts, not just commands.
- Know OSI, TCP/IP, and protocols well.
- Read questions carefully.
- Manage time (don't spend too long).
- Review and verify your answers.
- Practice, Practice, Practice!



FINAL TIP

Consistency + Practice + Patience
= CCNA Success!

Keep Learning,
Keep Growing!

11) IP (Internet Protocol)

Date : ____ / ____ / ____

Page No. : _____



IP (Internet Protocol) is a **connectionless** network layer protocol used to deliver packets from the source host to the destination host across multiple networks. It provides **logical addressing** and **routing** of packets.

★ Key Points :

- ✓ Works on Network Layer (Layer 3)
- ✓ Connectionless (No connection setup)
- ✓ Uses logical addressing (IP Address)
- ✓ Best-effort delivery (No guarantee of delivery)
- ✓ Responsible for routing and forwarding packets
- ✓ Supports fragmentation and reassembly
- ✓ Protocol Number : 4
- ✓ Used by TCP, UDP, ICMP, IGMP, etc.

★ IP Addressing :

- IP address is a 32-bit logical address.
- Written in dotted-decimal notation (IPv4)
Example : **192.168.1.10**
- Identifies the network and the host.

Classes of IPv4 Addresses

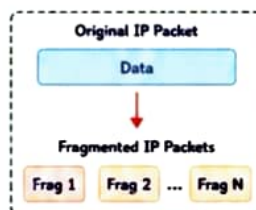
Class	Leading Bits	First Octet Range	Default Subnet Mask	No. of Networks	Hosts per Network
A	0	1 - 126	255.0.0.0	126	16,777,214
B	10	128 - 191	255.255.0.0	16,384	65,534
C	110	192 - 223	255.255.255.0	2,097,152	254
D	1110	224 - 239	N/A	Multicast	N/A
E	11110	240 - 255	N/A	Reserved	N/A

IP Packet (IPv4 Header Format)

0	8	16	24	31
Version(4)	IHL(4)	Type of Service (8)	Total Length (16)	
Identification (16)			Flags (3)	Fragment Offset (13)
Time to Live (8)		Protocol (8)	Header Checksum (16)	
Source IP Address (32)				
Destination IP Address (32)				
Options (0 to 40 bytes)				Padding
Data (Payload)				

★ IP Fragmentation and Reassembly

- If the packet size is greater than the MTU (Maximum Transmission Unit), the packet is fragmented.
- Each fragment has same Identification, different Fragment Offset.
- Reassembly is done at the destination host.



★ IP Routing

- ✓ IP uses routing tables to forward packets.
- ✓ Routers examine destination IP and forward the packet to the next best hop.
- ✓ Uses routing protocols (RIP, OSPF, BGP, etc.).



Uses of IP

- Internet communication
- Routing and forwarding
- Works as the backbone of internet
- Supports higher layer protocols

IP Communication



Characteristics of IP

1. Connectionless service
2. Logical addressing using IP address
3. Routing of packets across multiple networks
4. Best-effort delivery (No reliability)
5. Media independent
6. Supports fragmentation and reassembly
7. Used for unicast, broadcast and multicast

★ Types of IP Addresses :

- ✓ Unicast - One to One
- ✓ Broadcast - One to All (in a network)
- ✓ Multicast - One to Many (group)
- ✓ Anycast - One to Nearest

- Version : IP version (4 for IPv4)
- IHL : Internet Header Length
- Type of Service : QoS / Priority
- Total Length : Total datagram size
- Identification : Used for fragmentation
- Flags : Fragmentation control
- Fragment Offset : Position of fragment
- TTL : Time to Live (hop count)
- Protocol : Next layer protocol (TCP=6, UDP=17, ICMP=1)
- Header Checksum : Error check for header
- Source IP : 32-bit source address
- Destination IP : 32-bit destination address
- Options : Optional fields
- Padding : To make header multiple of 32 bits

IP (IPv4) vs IPv6

Feature	IPv4	IPv6
Address Size	32-bit	128-bit
Address Notation	Dotted Decimal	Hexadecimal
Header Size	20 - 60 bytes	40 bytes (Fixed)
Checksum	Header + Data	Header only
Broadcast	Supported	Not supported (Uses Multicast)
NAT	Required	Not required
Security (IPSec)	Optional	IPSec (Mandatory support)
Address Space	~4.9 Billion	~340 Undecillion

Additional IP Related Concepts

- ✓ Private IP Ranges (IPv4)
10.0.0.0 - 10.255.255.255 (Class A)
172.16.0.0 - 172.31.255.255 (Class B)
192.168.0.0 - 192.168.255.255 (Class C)
- ✓ Loopback Address : 127.0.0.1
- ✓ Default Gateway : Route to external network

Note :

💡 IP provides logical addressing and routing of packets across networks but does not guarantee delivery.



CCNA TOOLS

12 ESSENTIAL TOOLS EVERY CCNA STUDENT SHOULD KNOW



1 Cisco Packet Tracer



WHAT IT DOES

Network simulation tool by Cisco that allows you to design, build, configure and troubleshoot networks in a virtual environment.

USED FOR / EXAMPLES

- Practice device configuration (routers, switches, PCs)
- Test network scenarios and connectivity
- Learn before working on real devices



COMMON IOS COMMANDS (QUICK REFERENCE)

EXEC MODE COMMANDS

```
show ip interface brief
show interfaces
show version
show running-config
ping <ip-address>
traceroute <ip-address>
show ip route
```

GLOBAL CONFIGURATION MODE COMMANDS

```
hostname <name>
no ip domain-lookup
service password-encryption
banner motd <message>
clock set <hh:mm:ss> <day> <month> <year>
```

INTERFACE CONFIGURATION MODE COMMANDS

```
ip address <ip> <mask>
no shutdown
shutdown
description <text>
speed auto
duplex auto
```

SAVE CONFIGURATION

```
copy running-config
startup-config
(write memory)
```

TIPS

- ✓ Practice regularly in labs.
- ✓ Understand what each tool does and when to use it.
- ✓ Combine these tools for effective troubleshooting and network management.
- ✓ The more you lab, the stronger your CCNA foundation.

2 GNS3



WHAT IT DOES

Graphical Network Simulator that emulates real network devices using IOS images.

USED FOR / EXAMPLES

- Build complex network topologies
- Test routing protocols, VPN, firewalls
- Use real IOS for realistic labs

3 Wireshark



WHAT IT DOES

Network protocol analyzer that captures and analyzes packets in real time.

USED FOR / EXAMPLES

- Troubleshoot network issues
- Analyze protocols (TCP, UDP, DNS, HTTP, etc.)
- Verify communication between devices

4 PuTTY



WHAT IT DOES

SSH and Telnet client used to securely connect to network devices via CLI.

USED FOR / EXAMPLES

- Access routers/switches remotely
- Configure and manage devices
- Secure CLI access (SSH)

5 Tera Term



WHAT IT DOES

Terminal emulator used for connecting to network devices via serial or network.

USED FOR / EXAMPLES

- Access devices via console connection
- Configure settings (baud rate, terminal type)
- Useful for lab and real device access

6 SecureCRT



WHAT IT DOES

Advanced terminal emulator supporting SSH, Telnet, and serial connections.

USED FOR / EXAMPLES

- Secure CLI access
- Session management
- Automation with scripts

7 IP Calculator / Subnetting Tools



WHAT IT DOES

Calculates IP addresses, subnets, wildcard masks, and ranges.

USED FOR / EXAMPLES

- Subnet planning
- IP address calculations
- VLSM, CIDR, and subnet verification

8 Nmap



WHAT IT DOES

Network scanning tool used to discover hosts and services on a network.

USED FOR / EXAMPLES

- Host discovery
- Port scanning
- Network security assessment

9 SolarWinds Network Topology Mapper



WHAT IT DOES

Network mapping tool that discovers and visualizes your network topology.

USED FOR / EXAMPLES

- Automatic network discovery
- Topology mapping
- Monitor network performance

10 Notepad++ or VS Code



OR



WHAT IT DOES

Text editors used for writing configs, scripts, and notes.

USED FOR / EXAMPLES

- Edit configuration files
- Write Python automation scripts
- Take and organize lab notes

11 Cisco IOS Command Reference



WHAT IT DOES

Official Cisco command reference guide for all IOS commands.

USED FOR / EXAMPLES

- Find command syntax
- Understand command usage
- Reference for troubleshooting

12 Online Labs (DevNet / NetLab)



WHAT IT DOES

Cloud-based labs that provide real devices and sandboxes for practice.

USED FOR / EXAMPLES

- Hands-on practice on real devices
- Free labs and learning resources
- Build and test network solutions



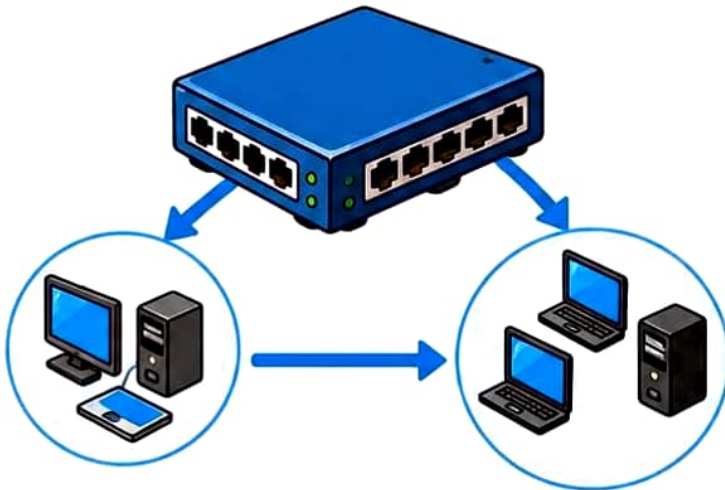
LAYER 2 SWITCH vs LAYER 3 SWITCH

www.thenetworkdna.com

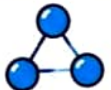
LAYER 2 SWITCH

DATA LINK LAYER

(OSI Layer 2)



- MAC Addresses
- Within Same VLAN
- MAC Address Table
- VLAC Address Table
- VLAN Segmentation
- No Inter-VLAN Routing

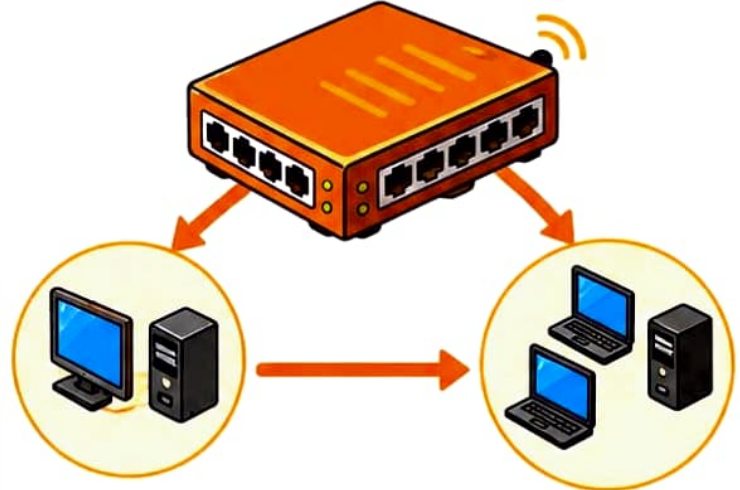


**LAYER 2 =
SAME NETWORK**

LAYER 3 SWITCH

NETWORK LAYER

(OSI Layer 3)



- IP Addresses
- Inter-VLAN Routing
- Routing Protocols (OSPF, EIGRP)
- Default Gateway
- Switching + Routing



**LAYER 3 =
DIFFERENT NETWORKS**